

EM-538 Machine Learning Project Proposal

LLM-Aided Customer Support Automation (Intent Detection)

1. Introduction

a. Problem Definition

Every day, customer service representatives deal with thousands of questions via social media, email, and chat. For businesses in the transportation, banking, telecommunications, and e-commerce sectors in particular, platforms such as Twitter (now X) have developed into the hubs for real-time client complaints. Quick and correct responses are crucial, but human agents find it difficult to keep up with high volume, inconsistent linguistic patterns, and customers' demands for immediate assistance.

This project combines machine learning (ML) and large language models (LLMs) to solve the issue of automatically comprehending customer complaints and producing suitable assistance responses. After determining the purpose of a customer's communication (such as a billing issue, delivery delay, or technical breakdown), the system must assess its severity and provide a tailored, brief, and expert answer.

The recommended strategy uses modern natural language processing (NLP) and generative artificial intelligence (AI) to handle noisy, real-world data written in social media style, in contrast to conventional rule-based chatbots that struggle with casual, cruel, or incomplete language. Accurately identifying user intent in brief communications and generating thoughtful responses that uphold company tone, empathy, and clarity are difficult tasks.

b. Significance

In businesses that require quick response times, automating customer service has grown in significance. The project is useful in a number of areas:

- **Operational Efficiency:** By filtering, categorizing, and automatically reacting to typical problems, businesses lessen the workload for human agents.
- **Customer satisfaction:** Quick, precise responses enhance the user experience, lessen annoyance, and boost brand confidence.
- **Cost Reduction:** In large support organizations, automated evaluation can save millions of dollars every year.
- **Scalability:** ML + LLM pipelines can handle millions of messages without requiring significant staffing adjustments.

Conventional customer support automation systems use manually curated rules or keyword matching. When users utilize slang, mistakes, sarcasm, or many issues in a single communication, these methods are readily broken. By combining transformer-based intent

detection with LLM reasoning and generation, this effort closes a significant gap by enabling the system to comprehend natural language patterns and produce relevant replies.

2. Objectives

a. Project Objectives

- Create a machine learning classifier with at least 80% accuracy in predicting customer complaint categories.
- Create a severity prediction model that assigns a 0–3 urgency score to complaints.
- Make an LLM-based response generator that generates succinct, sympathetic, and assistance-ready responses.
- Provide 2 to 3 sentence LLM explanations explaining the classifier's prediction for a particular category.
- Use accuracy, F1-score, confusion matrix, and human quality ratings to assess the entire system.

b. Expected Outcomes

- A working machine learning pipeline for classifying consumer tweets in real time.
- LLM-generated output improves support responses.
- A hybrid system that demonstrates contemporary ML + LLM integration appropriate for customer service in the real world.
- A small dataset of intent and severity descriptors that is semi-automated.

3. Data Collection

a. Dataset Source

This project uses the kaggle dataset- Customer support on twitter:

<https://www.kaggle.com/datasets/thoughtvector/customer-support-on-twitter>

It contains conversation pairs between customers and the companies such as AmazonHelp, UberSupport, and airline support accounts. Examples in the dataset include charges, delays, malfunctioning apps, account access problems and customer service issues.

b. Data Preprocessing Plan

- Normalization of punctuation and lowercasing.
- Eliminate special characters, URLs, mentions, and emoticons.
- Annotation for intent categories with LLM assistance.
- Severity labeling with LLM assistance (0–3).
- Tokenizing with the tokenizer from BERT.

This dataset is well suited for classification and to generate response due to it's size, real conversational style, and driver issues type.

4. Methods

a. Machine Learning Models

A tiered model approach will be used to carry out the categorization task, and they are as follows:

1. Baseline Model:

- Logistic Regression + TF-IDF.
- creates a baseline for lower-bound performance.

2. Deep Learning Model:

- LSTM or GRU.
- Captures sequential context and informal language patterns.

3. Transformer-Based Model:

- Fine-tuning DistilBERT or BERT
- Expected to provide the best accuracy for short-form text classification.
- used to forecast severity levels as well.

b. LLM Integration

LLMs will be tied together with ML models in three areas:

1. Response Generation:

An LLM generates the following given tweet text, predicted intent, and severity:

- Brief, sympathetic support responses.
- Tone changes that are optional (formal, short, friendly)

2. Explanations of classifications:

LLMs support model interpretability by producing brief explanations of why a tweet falls into a category.

3. Data Augmentation:

For underrepresented groups, LLMs generate artificial complaint cases to enhance model accuracy and balance.

Hyperparameter Tuning & Model Selection:

Grid search for the baseline classifiers, early stopping for the LSTM/GRU models to avoid overfitting, and learning-rate adjustments when fine-tuning BERT will all be used in model tuning. Accuracy, precision, recall, F1-score, and confusion matrices will be used to evaluate the system. Additionally, basic 1–5 human evaluations will be used to evaluate the quality of LLM-generated responses.